IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Name: Shehraz Sarwar
Date: 25/10/2025

Outline

- Introduction
- Executive Summary
- Methodology
- Results
- Conclusion
- Appendix

Project By: Shehraz Sarwar

Introduction

Space exploration has always been an expensive endeavor, but **SpaceX revolutionized the industry** by developing **reusable rockets**, drastically reducing the cost of launches.
While other providers charge over **$165 million** per launch, SpaceX offers the **Falcon 9** launch service for only **$62 million,** a difference largely due to the successful **landing and reuse of the first stage**.

Understanding **what factors influence a successful landing** can therefore help estimate the **true cost of a launch** and provide valuable insights to **competitors or clients bidding** for launch contracts.

The goal of this project is to **predict whether the Falcon 9 first stage will successfully land** using historical launch data from SpaceX. By analyzing, visualizing, and modeling real mission data, we aim to uncover **patterns that influence landing outcomes** and build a **machine learning model** to predict future success rates.

Project By: Shehraz Sarwar

Executive Summary

Summary of Methodologies:

- Gathered SpaceX data using API and Web Scraping.
- Cleaned and prepared data through Python-based wrangling.
- Performed EDA with SQL to answer key analytical questions.
- Conducted visual EDA in Python to explore payload and success patterns.
- Used Folium maps to visualize launch and landing sites.
- Built an interactive dashboard (Dash + Plotly) for dynamic analysis.
- Applied Machine Learning models to predict landing success.

Summary of Results:

- Key factors are payload mass, launch site, and booster version influence landing success.
- Dashboard Enabled easy comparison of site performance and payload-success correlation.
- Best Model is Decision Tree
- Cross-Validation Score is 0.916 (91.6%)
- Successful landings can be accurately predicted using historical data.

Section 1

Methodology

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Methodology

Executive Summary

1. **Data Collection:** Gathered SpaceX launch data using API and Web Scraping for comprehensive, up-to-date information.

2. **Data Wrangling (Python):** Cleaned, refined, and structured data by adding/dropping columns for analysis readiness.

3. **EDA (SQL):** Queried data to find unique launch sites, first successful landing, payload trends, and mission outcomes.

4. **EDA (Python):** Used Matplotlib and Seaborn to visualize payload effects, launch site success, and overall mission trends.

5. **Geospatial Analysis (Folium):** Created interactive maps to display launch and landing sites geographically.

6. **Dashboard (Dash + Plotly):** Built an interactive dashboard for viewing top sites, payload-success correlation, and mission insights.

7. **Machine Learning:** Trained and compared models to predict landing success.

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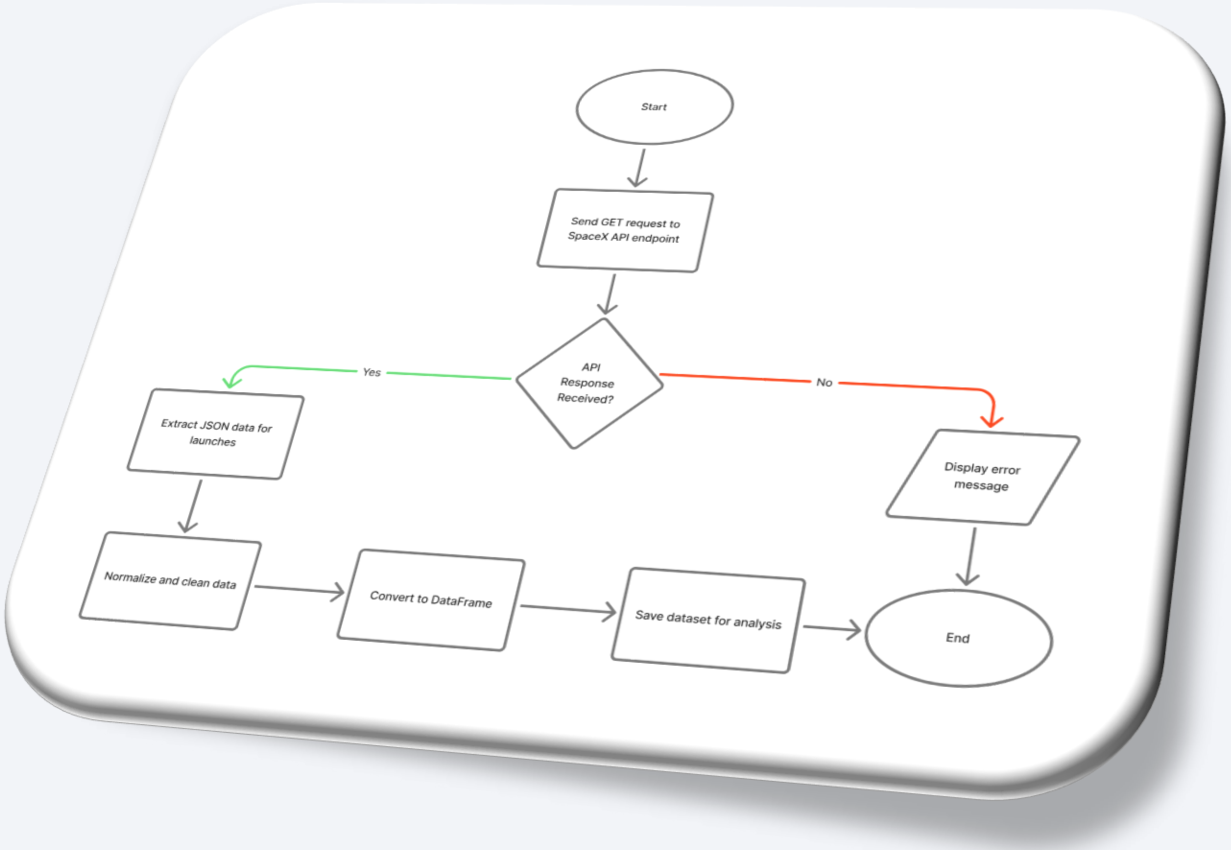
Data Collection – SpaceX API

Data Collection Process – SpaceX API

- Data collected using the **SpaceX REST API** (https://api.spacexdata.com/v4/launches/past)
- API responses parsed into **Pandas DataFrames** using requests and json_normalize()
- Additional details (rocket type, payload, launch site, core data) fetched through helper functions calling related API endpoints.
- Final dataset saved as **dataset_part_1.csv**
- GitHub Repository: SpaceX Launch Data Analysis – Part 1 (Data Collection)



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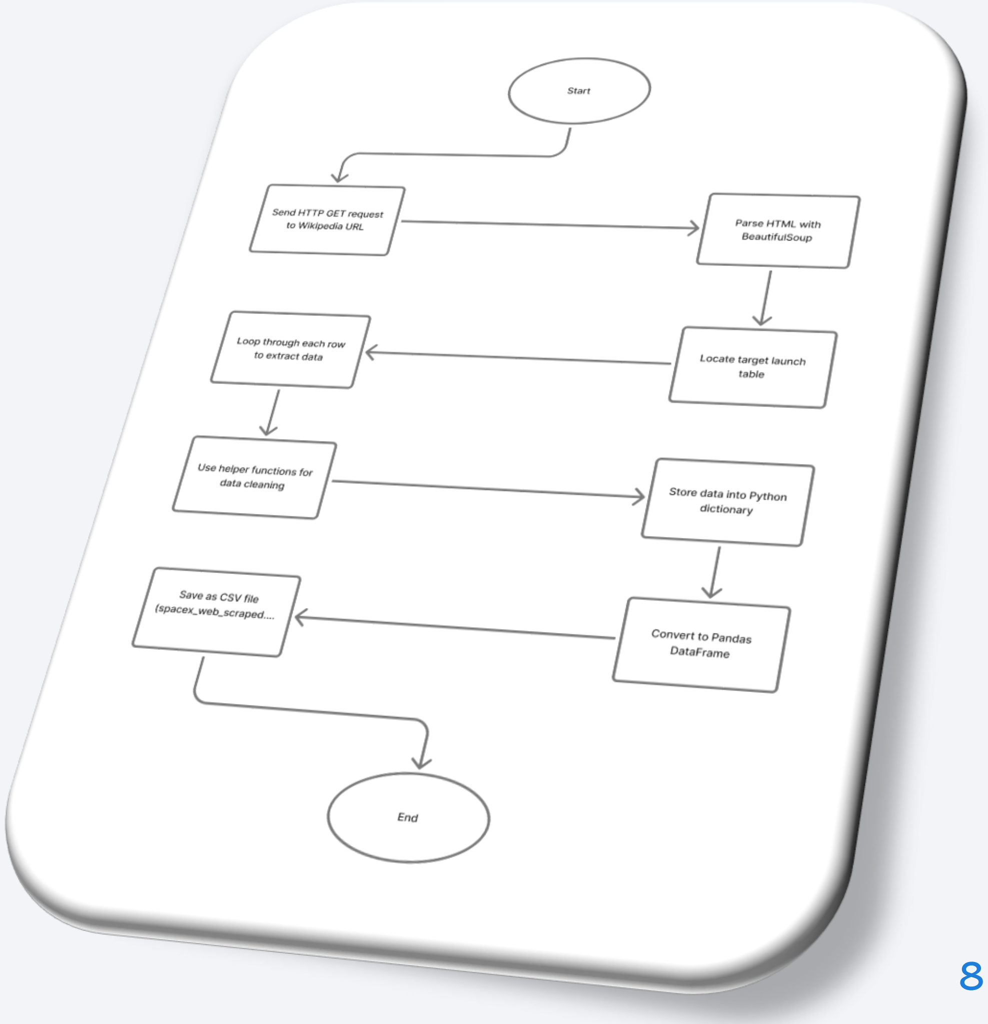
Data Collection - Scraping

Data Collection Process - Scraping

- Scraped data from the Wikipedia page using **Python's requests** and **BeautifulSoup** libraries.
- Extracted key details such as Flight No., Date, Time, Launch Site, Payload, Mass, Orbit, Customer, Outcome, and Landing Status.
- Implemented **custom helper functions** to clean and standardize columns from raw HTML tables.
- Parsed and structured all data into a Pandas DataFrame, then saved as a **CSV file.**
- GitHub Repository: [SpaceX Launch Data Analysis – Part 1 (Data Collection)]



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Data Wrangling

1. Data Import
- Loaded SpaceX launch dataset directly from cloud storage (CSV).

2. Missing Value Analysis
- Identified missing values (e.g., *LandingPad – 28.88% missing*).
- Checked completeness for all attributes using .isnull().

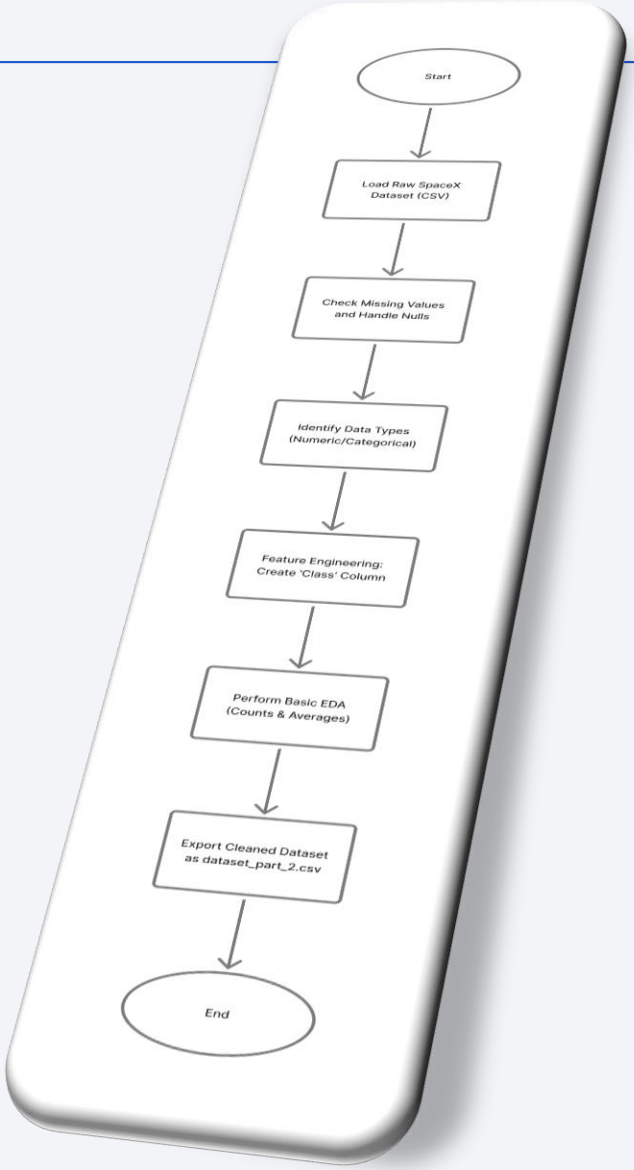
3. Data Type Classification
- Separated **numerical** and **categorical** columns using df.dtypes.
- Ensured proper type consistency for modeling.

4. Feature Engineering
- Created new column **Class** to indicate landing success (1 = success, 0 = failure).

5. Exploratory Insights
- Counted launches per site and orbit using .value_counts().
- Determined average landing success rate: **67%**.

6. Data Export
- Final cleaned dataset saved as dataset_part_2.csv for modeling and visualization.

GitHub Repository: [SpaceX Launch Data Analysis – Part 2 (Data Wrangling)]



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EDA with SQL

Summary Of Tasks Performed & Insights

- **Task 1:** Retrieved all unique launch site names. (*Found:* CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40)
- **Task 2:** Filtered launch sites starting with "CCA". (*Used to analyze Cape Canaveral–based launches)*
- **Task 3:** Calculated total payload for NASA (CRS) missions. (*Total payload:* **45,596 kg**)
- **Task 4:** Computed average payload for Booster Version F9 v1.1. (*Average payload:* **2,928 kg**)
- **Task 5:** Found the first successful ground-pad landing date. (*Result:* **2015-12-22**)
- **Task 6:** Listed drone-ship successes with payloads 4,000–6,000 kg. (*Boosters:* F9 FT B1022, B1026, B1021.2, B1031.2)
- **Task 7:** Counted mission outcomes. (*100 successful / 1 failed)*
- **Task 8:** Used subquery to find boosters with maximum payload. (*Highest capacity:* **F9 Block 5 series**)
- **Task 9:** Extracted 2015 failed drone-ship landings by month and site. (*Failures in Jan & Apr 2015 – CCAFS LC-40)*
- **Task 10:** Ranked landing outcomes (2010–2017). (*Top outcomes:* **No Attempt > Success > Failure**)

GitHub Repository: [SpaceX Launch Data Analysis – Part 3 (Exploratory Data Analysis using SQL)]

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EDA with Data Visualization

Charts Plotted & Their Purpose

Chart Type	Variables	Purpose / Insight
Scatter Plot (catplot)	FlightNumber vs. PayloadMass (hue=Class)	To identify how payload mass and flight number relate to landing success.
Scatter Plot (catplot)	FlightNumber vs. LaunchSite	To observe launch site performance patterns over flight numbers.
Scatter Plot (catplot)	PayloadMass vs. LaunchSite	To detect payload distribution trends by launch site.
Bar Chart	Orbit vs. Success Rate (Class mean)	To compare landing success rates across different orbit types.
Scatter Plot (catplot)	FlightNumber vs. Orbit	To analyze the impact of flight experience on orbit-specific success.
Scatter Plot (catplot)	PayloadMass vs. Orbit	To study how payload affects success rate in different orbits.
Line Chart	Year vs. Success Rate	To visualize yearly trends in SpaceX's launch success progression.

GitHub Repository: [SpaceX Launch Data Analysis – Part 4 (Python EDA with Visualization)]

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Build an Interactive Map with Folium

Objects Added and Purpose

Markers and Circles (Launch Sites):
Added folium.Circle and folium.Marker for each launch site using latitude and longitude.
Purpose: To visualize launch site locations on the world map for better geographic context.

Colored Markers for Launch Outcomes:
Used green markers for successful launches (class=1) and red markers for failed ones (class=0).
Purpose: To easily observe which sites had higher success rates.

Distance Markers and PolyLines:
Created distance markers and folium.PolyLine objects connecting launch sites to nearby coastlines, cities, railways, and highways.
Purpose: To analyze how launch site proximity to infrastructure or coastlines might affect launch success or safety.

MousePosition Tool:
Added MousePosition control to get live latitude and longitude by hovering.
Purpose: To identify coordinates for nearby features (coastlines, cities, etc.) interactively.

GitHub Repository: [SpaceX Launch Data Analysis – Part 5 (Interactive Visuals with Folium)]

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Build a Dashboard with Plotly Dash

Dashboard Summary

Pie Chart:
Displays launch success counts. When “All Sites” is selected, it shows the total successful launches by site.
When a specific site is chosen, it shows **success vs failure rates** for that site.
Helps identify which launch sites have the highest success ratio.

Scatter Plot:
Plots **Payload Mass (kg)** against **Launch Outcome (Success/Failure)**, with points colored by **Booster Version Category**.
Helps analyze how payload size influences success and how different boosters perform.

Dropdown Menu (Launch Site):
Allows selection between all launch sites or a specific one.
Enables focused exploration of individual site performance.

Payload Range Slider:
Filters payload range dynamically for the scatter plot.
Helps zoom into specific payload intervals to observe trends.

GitHub Repository: SpaceX Launch Data Analysis – Part 6 (Interactive Dashboard with Plotly Dash)

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Predictive Analysis (Classification)

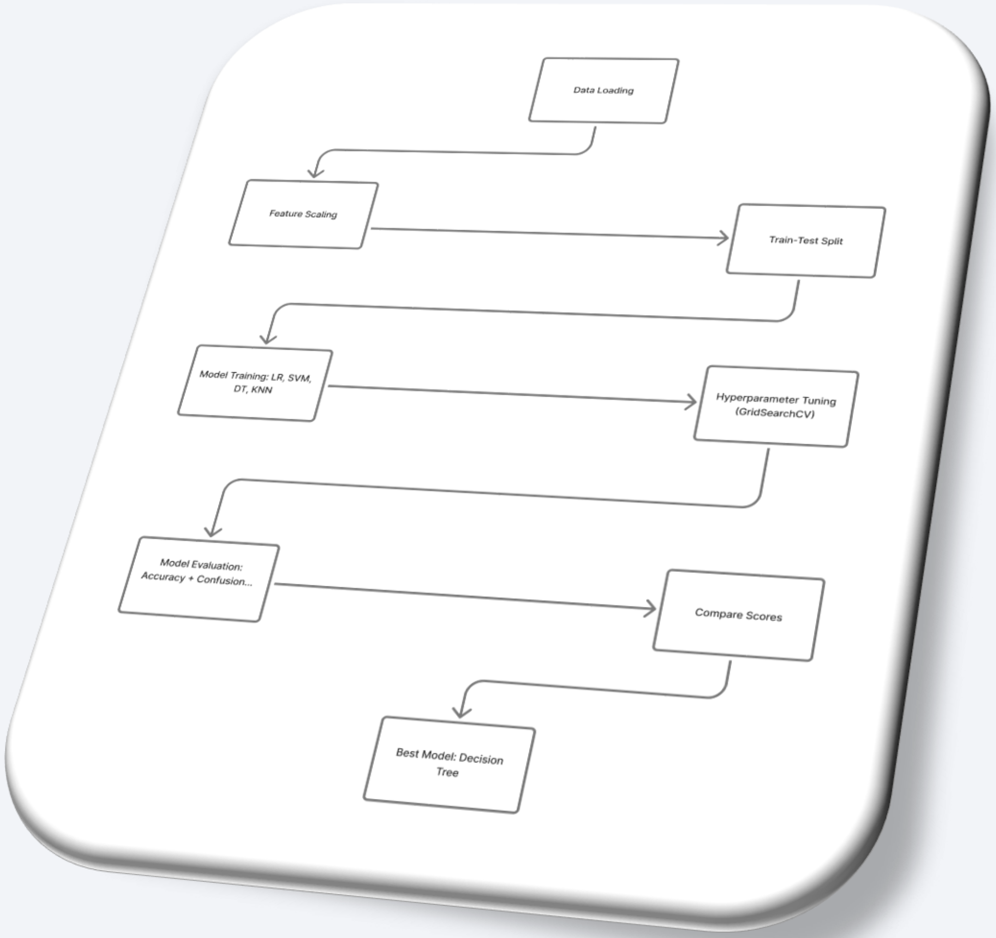
Model Development Summary

- **Data Preparation:** Loaded datasets, standardized features using StandardScaler(), and split data into 80% training & 20% testing sets.

- **Model Training:** Applied four models — Logistic Regression, SVM, Decision Tree, and KNN. Used **GridSearchCV (cv=10)** for hyperparameter tuning.

- **Performance:**
  - Logistic Regression → 84.6%
  - SVM → 84.8%
  - Decision Tree → **91.6%**
  - KNN → 84.8%

- **Evaluation:** All models achieved ~83% on test data; Decision Tree performed best in both validation and testing.



GitHub Repository: [SpaceX Launch Data Analysis – Part 7 (Predictive Analysis)]

Section 2

Insights drawn from EDA

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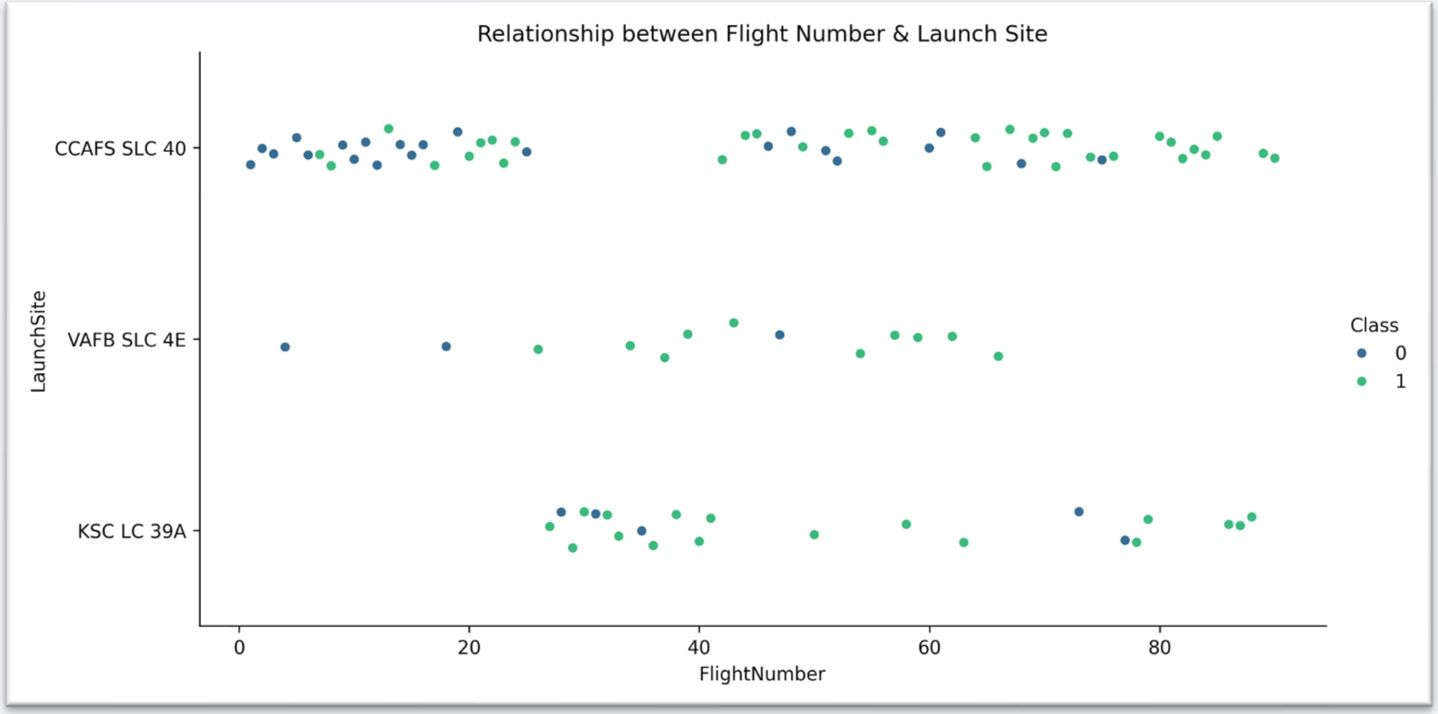
Flight Number vs. Launch Site

This plot shows the relationship between **Flight Number** and **Launch Site**, with each point representing a SpaceX launch colored by its outcome (blue for failure and green for success). The trend indicates that earlier flights, especially from CCAFS SLC 40, had more failures, while later flights across all sites show higher success rates. This suggests that SpaceX's launch reliability improved significantly over time as experience and technology advanced.



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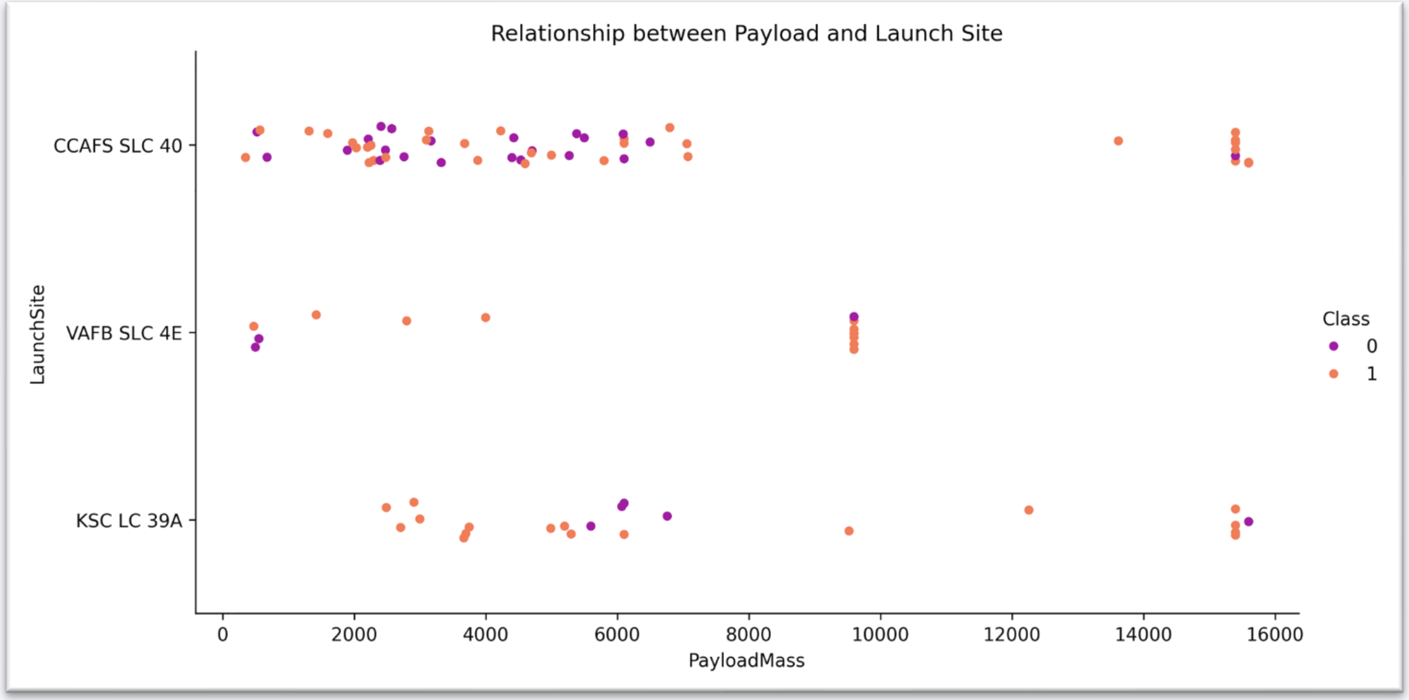
Payload vs. Launch Site

This plot illustrates the relationship between **Payload Mass** and **Launch Site**, with each point representing a SpaceX launch outcome (purple for failure and orange for success). It shows that all three launch sites CCAFS SLC 40, VAFB SLC 4E, and KSC LC 39A handled a wide range of payload masses, but the **success rate (orange points)** remains high across most payloads. Heavier payloads, especially above 10,000 kg, are mostly successful, indicating that **payload mass doesn't significantly affect launch success**, and the VAFB-SLC launch site there are no rockets launched for heavy payload mass(greater than 10000).



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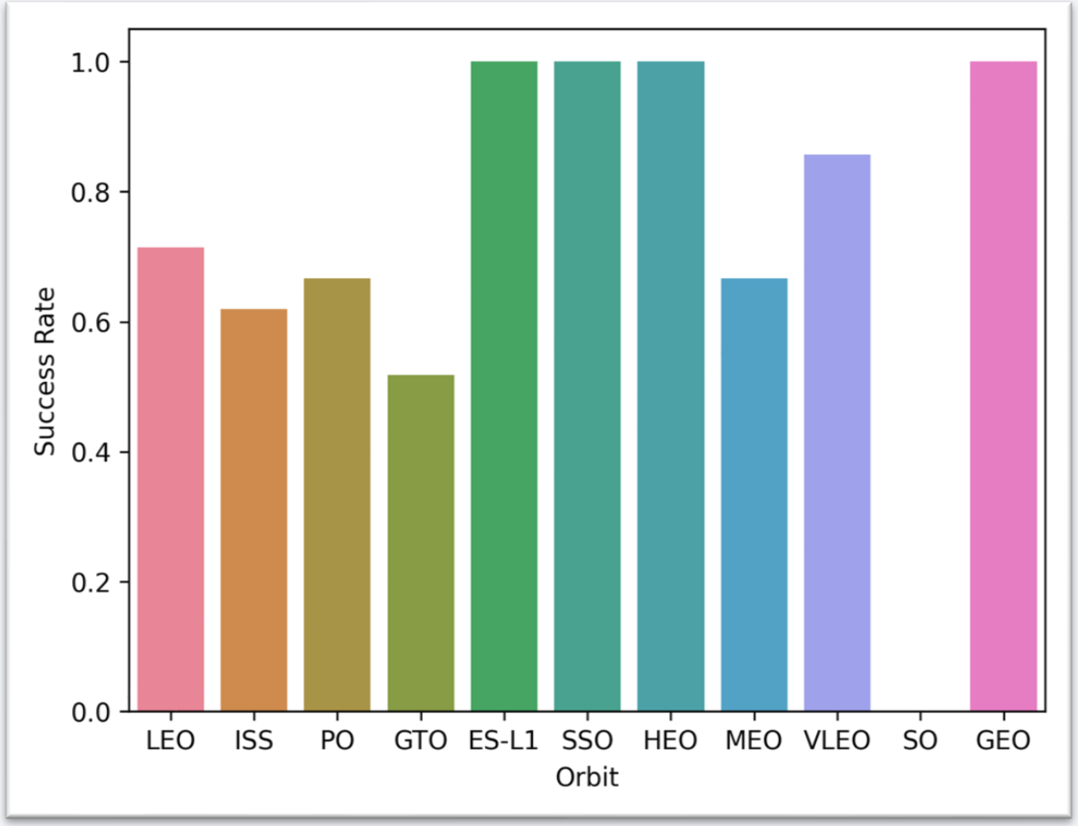
Success Rate vs. Orbit Type

This bar chart shows the **average launch success rate across different orbit types**. Each bar represents a specific orbit, such as **LEO (Low Earth Orbit)**, **ISS (International Space Station)**, **GTO (Geostationary Transfer Orbit)**, and others.

From the chart, we can see that **SSO (Sun-Synchronous Orbit)**, **HEO (Highly Elliptical Orbit)**, **ES-L1**, and **GEO (Geostationary Orbit)** have the **highest success rates (100%)**, indicating strong reliability for these missions. In contrast, **GTO** and **ISS** orbits show relatively lower success rates, suggesting these mission types faced more challenges and the **SO (Sub-Orbital)** orbit has a **0% success rate** meaning **all launches targeting SO orbit failed**.

Overall, the chart demonstrates that while early test orbits like **SO** had no success, operational orbits used in real missions achieved **high reliability** over time.



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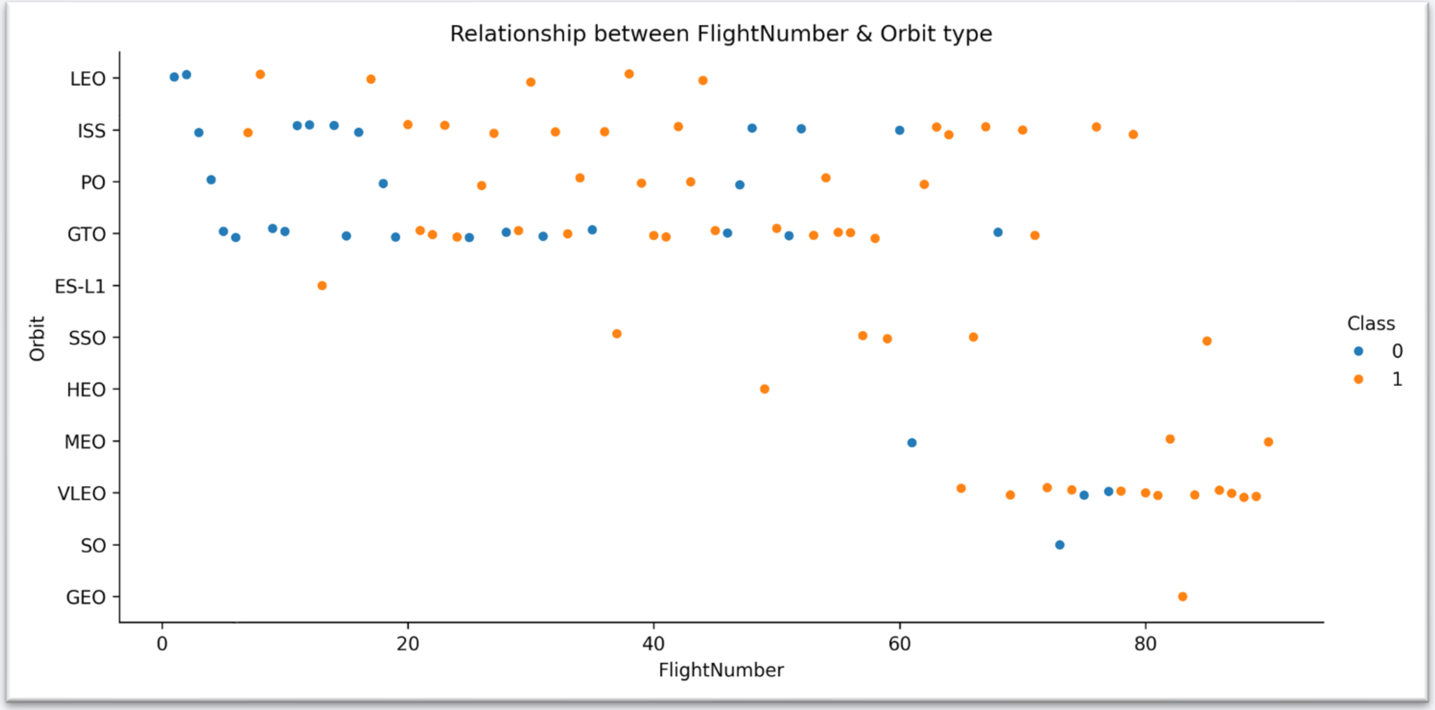
Flight Number vs. Orbit Type

The scatter plot illustrates the relationship between **Flight Number** and **Orbit type**, with each point representing a rocket launch classified by success (orange) or failure (blue). It shows that launches target a variety of orbits such as LEO, ISS, and GTO across different flight numbers, with higher flight numbers generally corresponding to more successful outcomes. This trend suggests that as the number of flights increased, the success rate improved, reflecting growing experience and reliability in launch performance over time.



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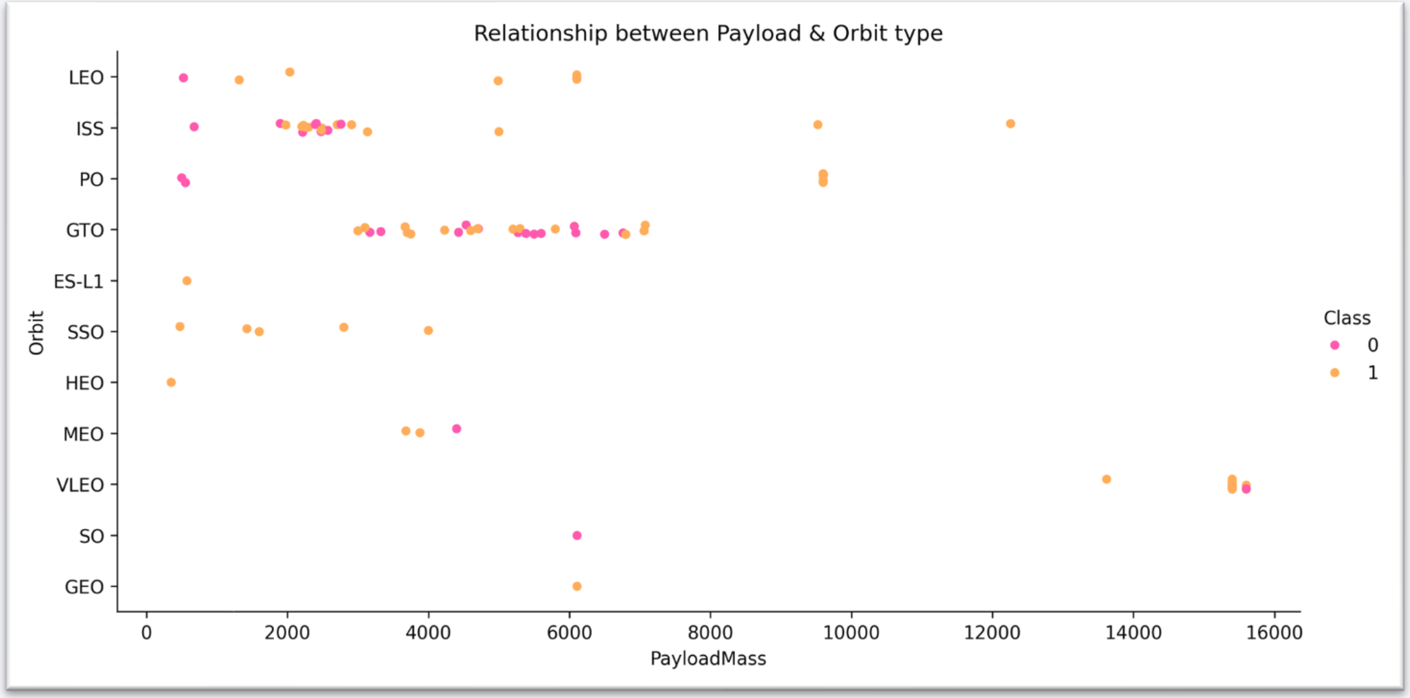
Payload vs. Orbit Type

This scatter plot shows the **relationship between Payload Mass and Orbit type**, with each point representing a rocket launch and color indicating the launch outcome (orange for success and pink for failure). The plot reveals that most payloads targeting **LEO**, **ISS**, and **GTO** orbits fall within the lower to mid payload mass range, while heavier payloads are associated with **VLEO** missions. Overall, successful launches (orange points) dominate across all orbit types, suggesting that **payload mass does not strongly affect launch success**, and rockets have become capable of handling diverse payload sizes across different orbital missions.



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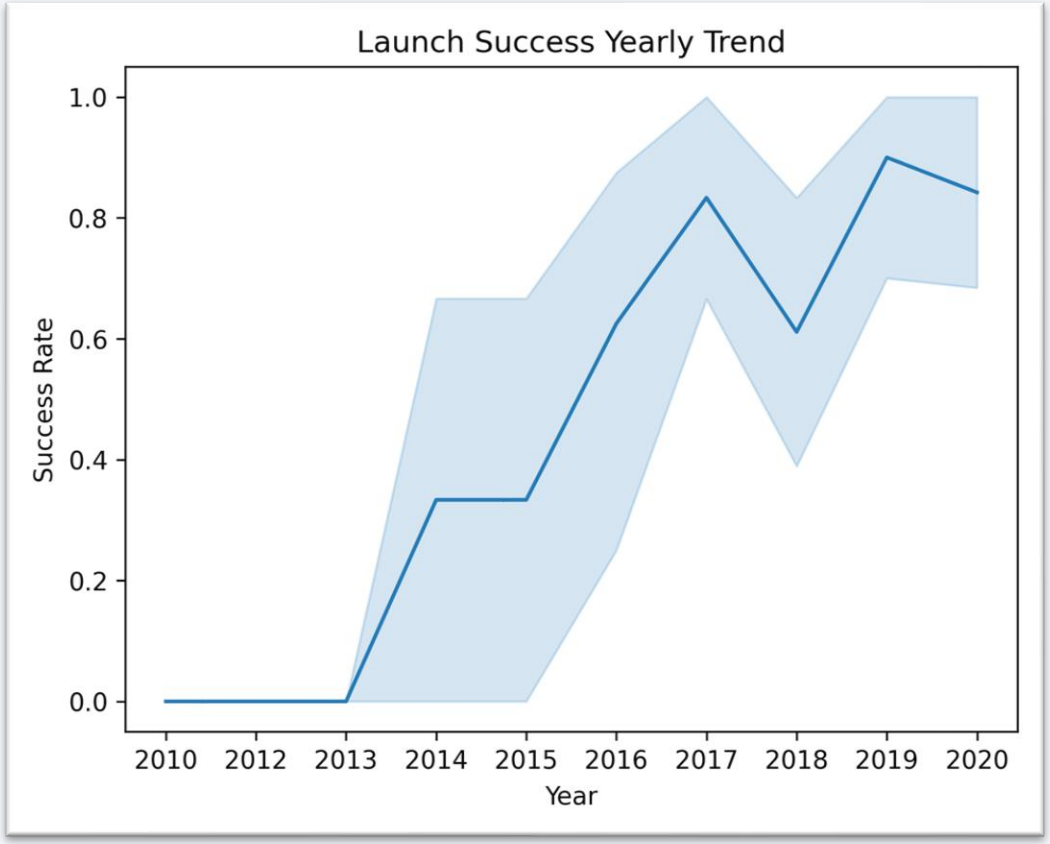
Launch Success Yearly Trend

This line chart illustrates the yearly success trend, showing that SpaceX's success rate remained steady in **2014.** However, starting in **2015**, the success rate began to **rise steadily**, showcasing significant advancements in engineering, mission planning, and reusability technologies.

By **2017**, SpaceX achieved a strong and consistent performance level, reflecting how the company's growing experience and innovations transformed early challenges into sustained reliability and mission success.



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All Launch Site Names

This query retrieves the **distinct launch site names** from the SpaceX dataset. It helps identify the different locations used for rocket launches, showing that missions were conducted from **CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, and CCAFS SLC-40**, representing the main operational launch sites.

In [14]:

```
%%sql
SELECT DISTINCT Launch_Site
FROM SPACEXTABLE
```

```
 * sqlite:///my_data1.db
Done.
```

Out[14]:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

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Launch Site Names Begin with 'CCA'

This query filters and displays **five records** where the launch site name begins with **‘CCA’**, using the LIKE operator. It demonstrates how pattern matching can be used in SQL to extract site-specific data, particularly focusing on **Cape Canaveral Air Force Station (CCAFS)** launch missions.

```
%%sql
SELECT * FROM SPACEXTABLE
WHERE Launch_Site LIKE 'CCA%'
LIMIT 5
```

```
* sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

This query calculates the **total payload mass** launched by the customer **NASA (CRS)** using the SUM() function. The result shows that NASA's CRS missions carried a total of **45,596 kg**, providing insight into the scale of payloads handled in these resupply missions.

```
%%sql
SELECT
    Customer,
    SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass
FROM SPACEXTABLE
WHERE Customer = 'NASA (CRS)'
GROUP BY Customer
```

```
 * sqlite:///my_data1.db
Done.
```

Customer	Total_Payload_Mass
NASA (CRS)	45596

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Average Payload Mass by F9 v1.1

This query determines the **average payload mass** for the **F9 v1.1 booster version** using the AVG() function. The calculated value of **2,928.4 kg** highlights the typical payload capacity of this specific booster model during its operational missions.

```
%%sql
SELECT
    Booster_Version,
    AVG(PAYLOAD_MASS__KG_) AS Avg_Payload_Mass
FROM SPACEXTABLE
WHERE Booster_Version = 'F9 v1.1'
GROUP BY Booster_Version
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version	Avg_Payload_Mass
F9 v1.1	2928.4

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First Successful Ground Landing Date

Using the MIN() function, this query finds the **earliest date** when a **successful ground pad landing** occurred. The result, **2015-12-22**, marks a significant milestone for SpaceX as it represents the **first-ever successful ground landing date**.

```
%%sql
SELECT
    MIN(Date) AS FirstDateSuccess
FROM SPACEXTABLE
WHERE Landing_Outcome LIKE '%ground pad%' AND
Mission_Outcome = 'Success'
```

```
 * sqlite:///my_data1.db
Done.
```

FirstDateSuccess
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

This query lists all **booster versions** that had **successful landings on drone ships** and carried payloads **between 4000 and 6000 kg**. It identifies boosters such as **F9 FT B1022, B1026, and B1031.2**, showcasing missions with both moderate payload mass and successful recovery performance.

```
%%sql
SELECT
    Booster_Version
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
AND Mission_Outcome = 'Success'
AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

This query groups all records based on **mission outcome**, categorizing them as either **Success** or **Failure**. The results reveal that out of all missions, **100 were successful and 1 failed**, reflecting SpaceX's **high success rate** across its flight history.

```
%%sql
SELECT
    CASE
        WHEN Mission_Outcome LIKE 'Success%' THEN 'Success'
        ELSE 'Failure'
    END AS Mission_Outcome_,
    COUNT(*) AS Total_Success
FROM SPACEXTABLE
GROUP BY CASE
            WHEN Mission_Outcome LIKE 'Success%' THEN 'Success'
            ELSE 'Failure'
         END
```

```
* sqlite:///my_data1.db
one.
```

Mission_Outcome_	Total_Success
Failure	1
Success	100

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Boosters Carried Maximum Payload

This query identifies all **booster versions** that carried the **maximum recorded payload mass** by using a **subquery with the MAX() function**.

The result lists multiple **F9 B5 boosters**, emphasizing that this advanced version handled the **heaviest payloads** in SpaceX missions.

```
%%sql
SELECT
    Booster_Version
FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE)
```

```
* sqlite:///my_data1.db
one.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

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2015 Launch Records

This query extracts the **month**, **landing outcome**, **booster version**, and **launch site** for missions in **2015** that had **failed landings on drone ships**. It shows that such failures occurred in **January and April 2015**, helping analyze early landing challenges during SpaceX's recovery phase.

```
%%sql
SELECT
    substr(Date, 6,2) AS Month_,
    Landing_Outcome,
    Booster_Version,
    launch_site
FROM SPACEXTABLE
WHERE Landing_Outcome LIKE 'Failure (drone ship)'
AND substr(Date,0,5)='2015'
```

```
 * sqlite:///my_data1.db
Done.
```

Month_	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Using DENSE_RANK() with aggregation, this query ranks different **landing outcomes** between **2010 and 2017** based on their frequency.

The results show that **‘No attempt’** was the most common outcome, followed by both **successful and failed drone ship landings**, providing a clear overview of landing performance trends over time.

```
%%sql
SELECT
    Landing_Outcome,
    COUNT(Landing_Outcome) AS Count_Landing_Outcome,
    DENSE_RANK() OVER (ORDER BY COUNT(Landing_Outcome) DESC) AS Rank_Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
```

```
* sqlite:///my_data1.db
one.
```

Landing_Outcome	Count_Landing_Outcome	Rank_Count
No attempt	10	1
Success (drone ship)	5	2
Failure (drone ship)	5	2
Success (ground pad)	3	3
Controlled (ocean)	3	3
Uncontrolled (ocean)	2	4
Failure (parachute)	2	4
Precluded (drone ship)	1	5

Section 3

Launch Sites Proximities Analysis

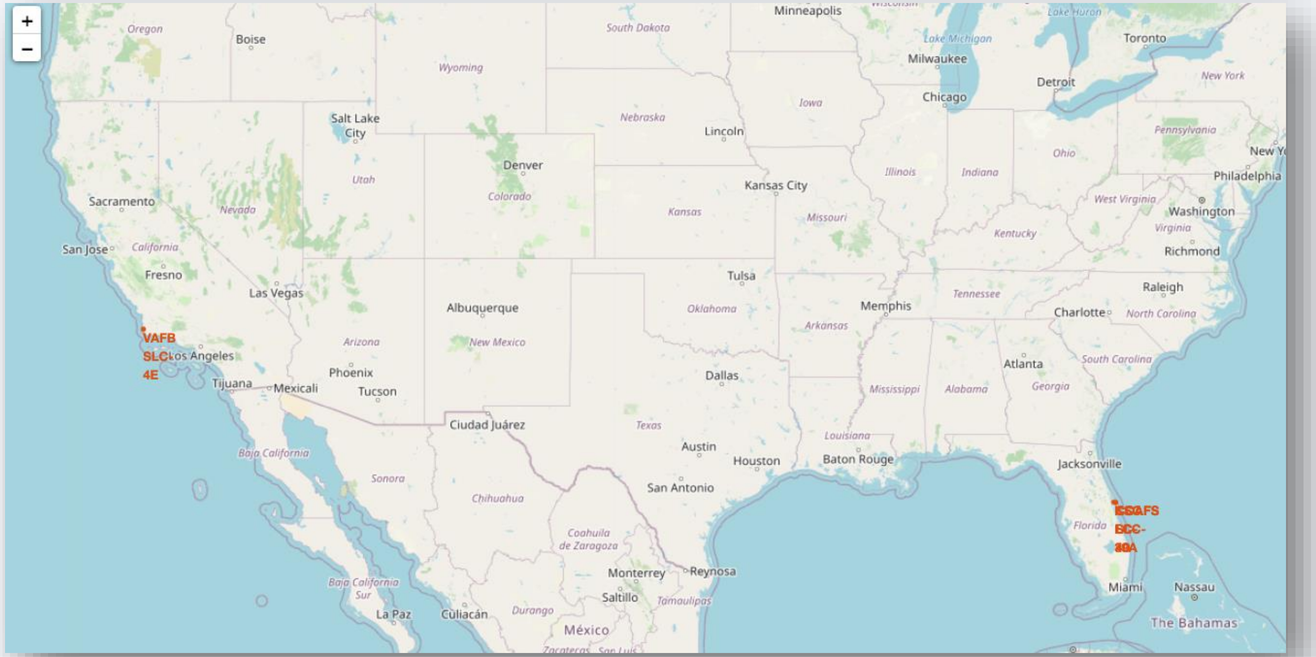
Project By: Shehraz Sarwar

Launch Site Locations Map

This map displays all SpaceX launch sites marked on the world map using their geographic coordinates (latitude and longitude).

What It Shows:

- Four launch sites: **CCAFS LC-40**, **CCAFS SLC-40**, **KSC LC-39A**, and **VAFB SLC-4E**.
- Each site is marked with a circle and labeled for easy identification.
- All sites are located **near the coastline** and relatively **close to the equator**, ideal for rocket launches.



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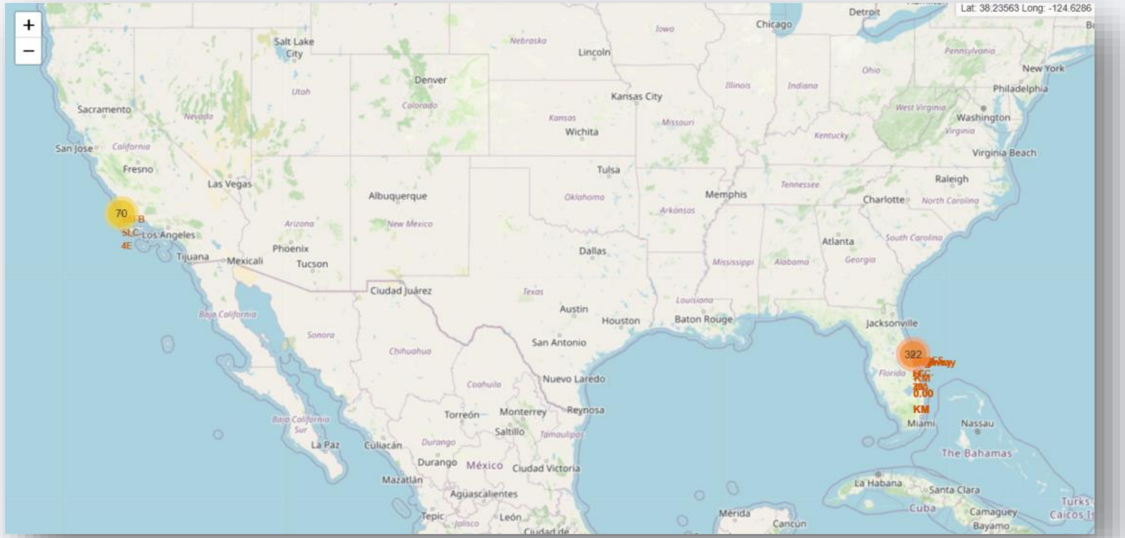
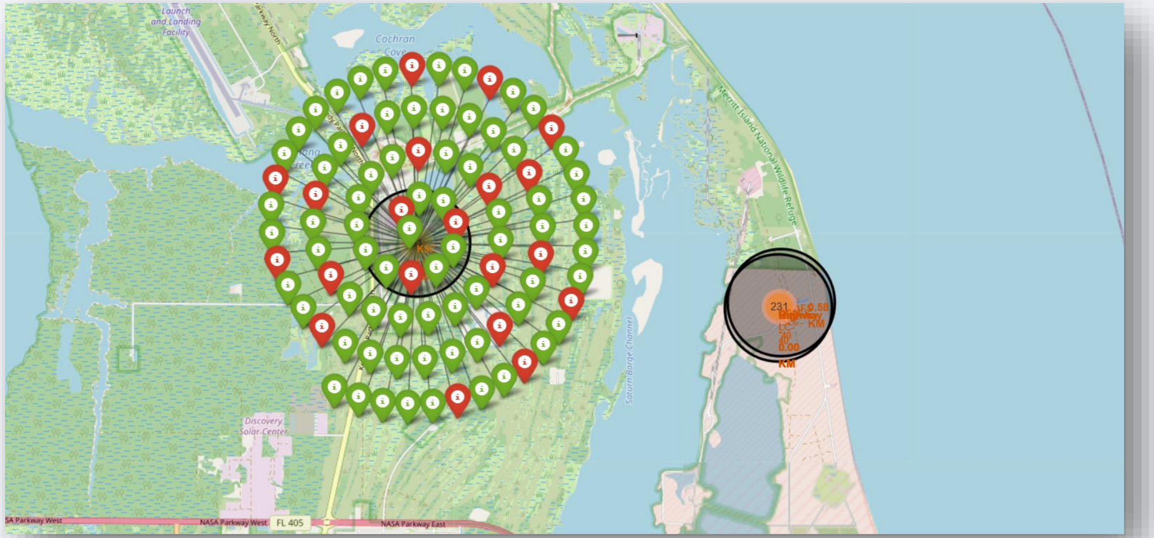
Launch Outcomes Map (Success vs Failure)

This map visualizes individual launch outcomes across all launch sites using color-coded markers.

What It Shows:

- **Green markers** → Successful launches
- **Red markers** → Failed launches
- **Clustered markers** indicate multiple launches from the same site.





Distance Analysis Map

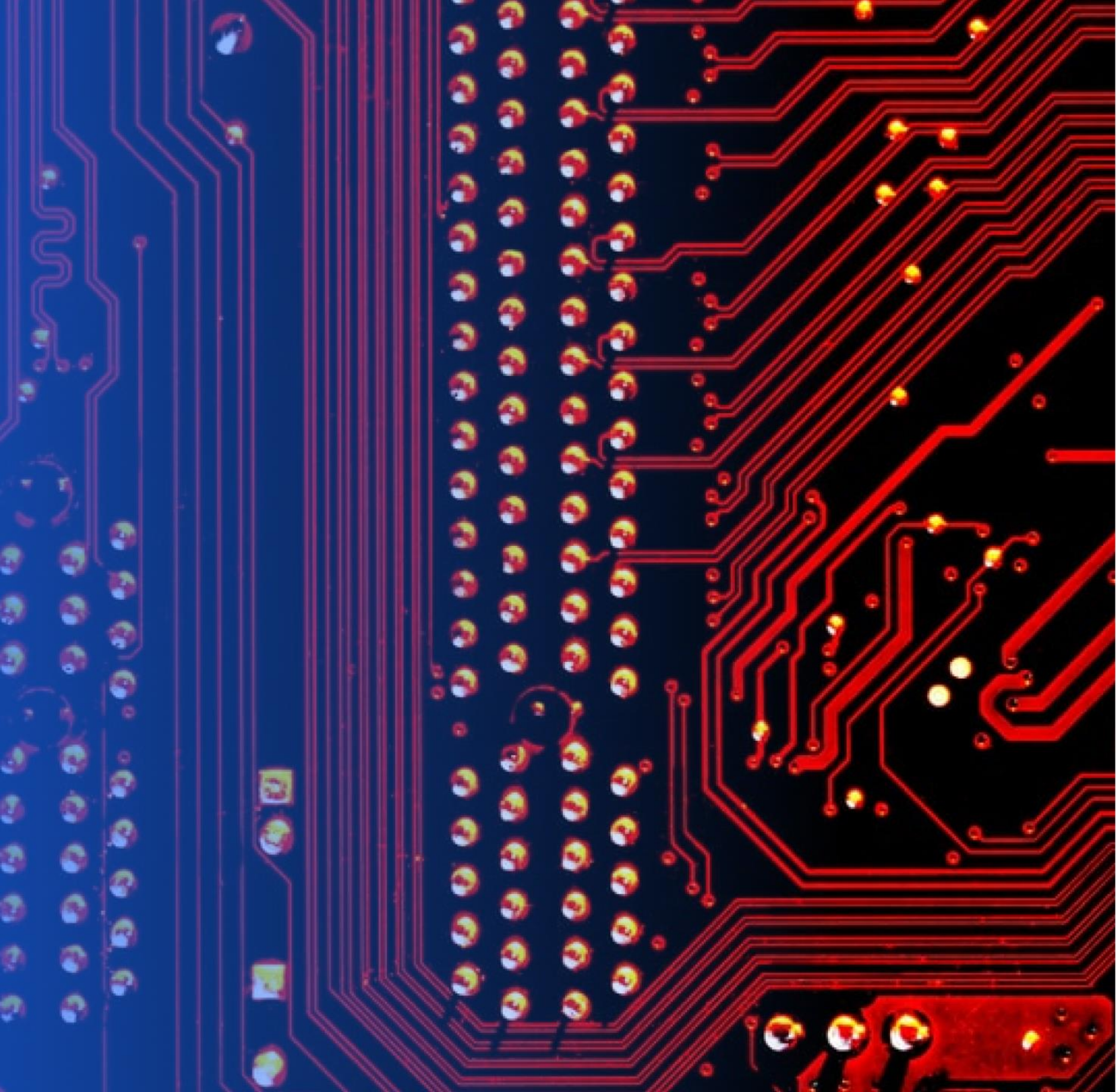
This map analyzes the **distance between launch sites and nearby features** like coastlines, railways, highways, and cities.

What It Shows:

- **Lines and labels** display distances (in KM) between launch sites and their nearest coastlines or infrastructure.
- Example: Distance from CCAFS LC-40 to coastline ≈ **0.58 KM**.

Section 4

Build a Dashboard with Plotly Dash



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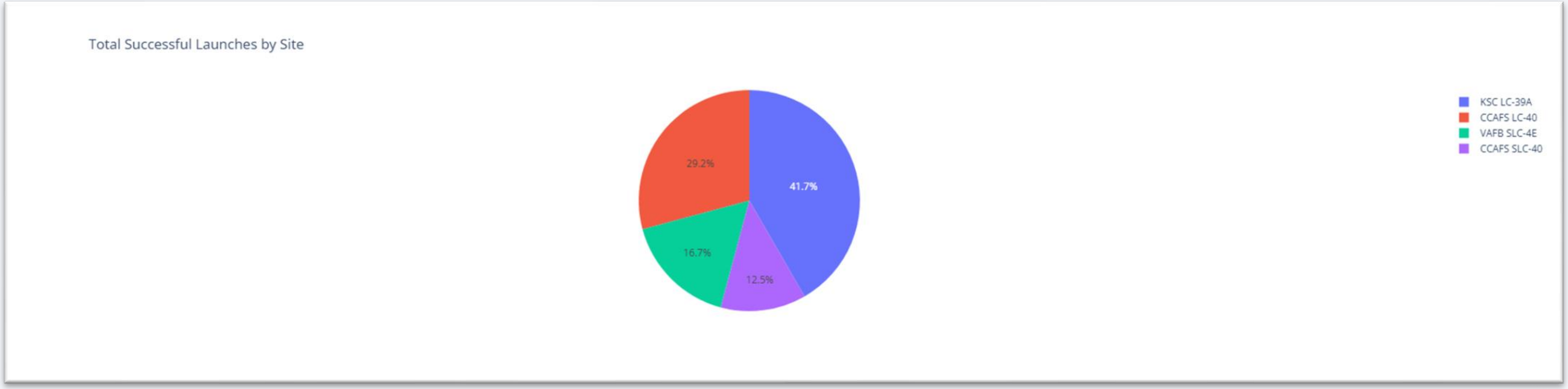
Success Rate of SpaceX Launches by Site

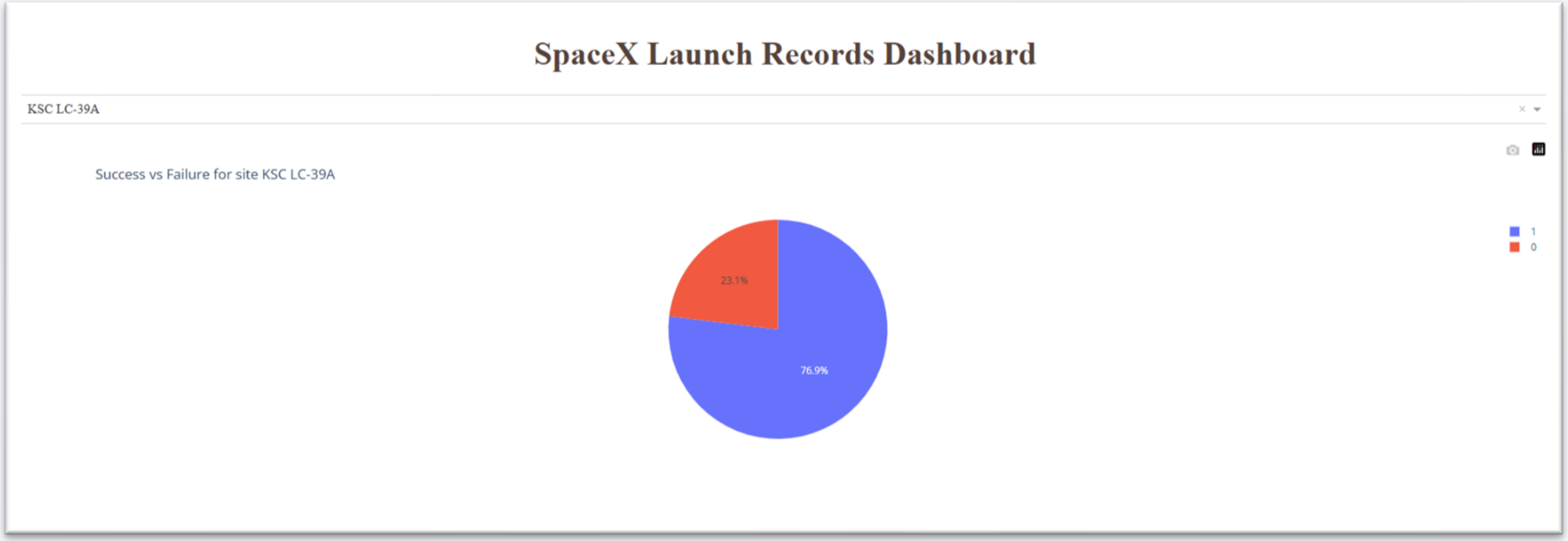
This pie chart illustrates the **total number of successful SpaceX launches by launch site**. Each segment represents a launch location and its percentage share of successful missions. The chart shows that **KSC LC-39A** accounts for the largest proportion of successful launches at **41.7%**, followed by **CCAFS LC-40 (29.2%)**, **VAFB SLC-4E (16.7%)**, and **CCAFS SLC-40 (12.5%)**. This indicates that the **Kennedy Space Center (KSC LC-39A)** has been the most frequently used and successful launch site among all SpaceX facilities.



Project By: Shehraz Sarwar

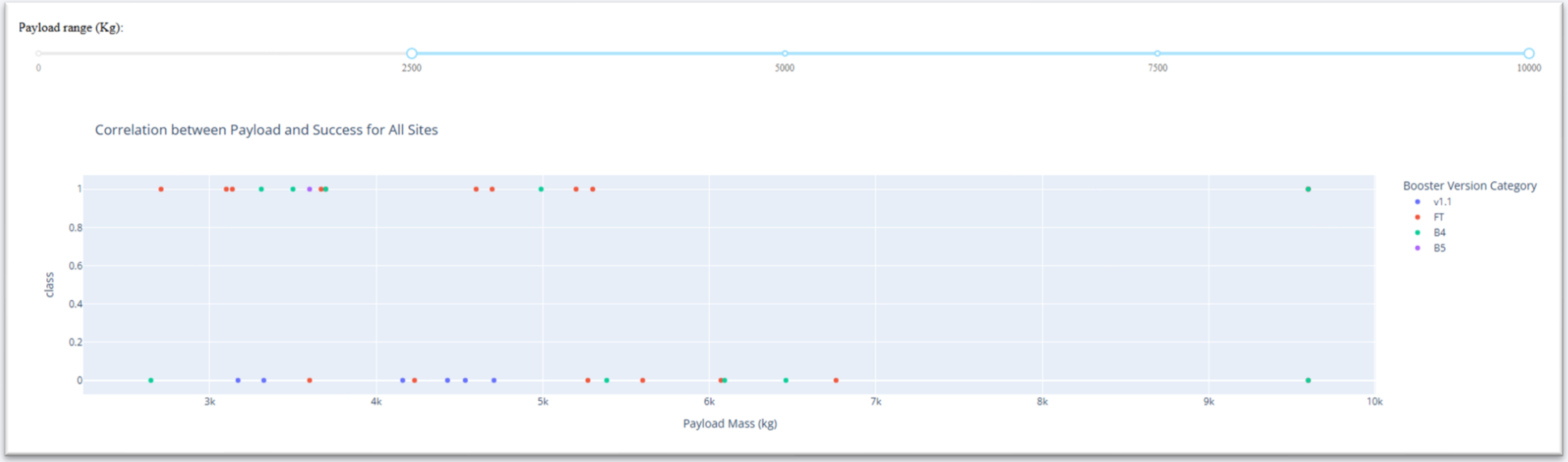
SpaceX's Most Successful Launch Site

The dashboard allows users to **analyze the distribution of successful and failed launches for each SpaceX launch site** using an interactive dropdown menu. By selecting a specific site, such as **KSC LC-39A**, the chart dynamically updates to show its performance metrics. **KSC LC-39A** the most successful site demonstrates a **76.9% success rate** and a **23.1% failure rate**, highlighting its strong reliability compared to other launch locations. This interactive feature helps users easily explore and compare **success-to-failure ratios** across different launch sites.



Correlation Between Payload Mass and Launch Success

The dashboard includes a **range slider** that allows users to **filter launches based on payload mass (in kilograms)**, enabling a focused analysis of how payload size affects launch success. The accompanying **scatter plot** visualizes the **correlation between payload mass and mission success** for either **all launch sites** or a **selected site** from the dropdown menu. Each point on the plot represents a launch, color-coded by **booster version category**, helping to identify performance trends across different rocket versions. This interactive visualization allows users to explore how **payload weight influences launch outcomes**, providing deeper insights into SpaceX's operational efficiency and success patterns.



Section 5

Predictive Analysis (Classification)

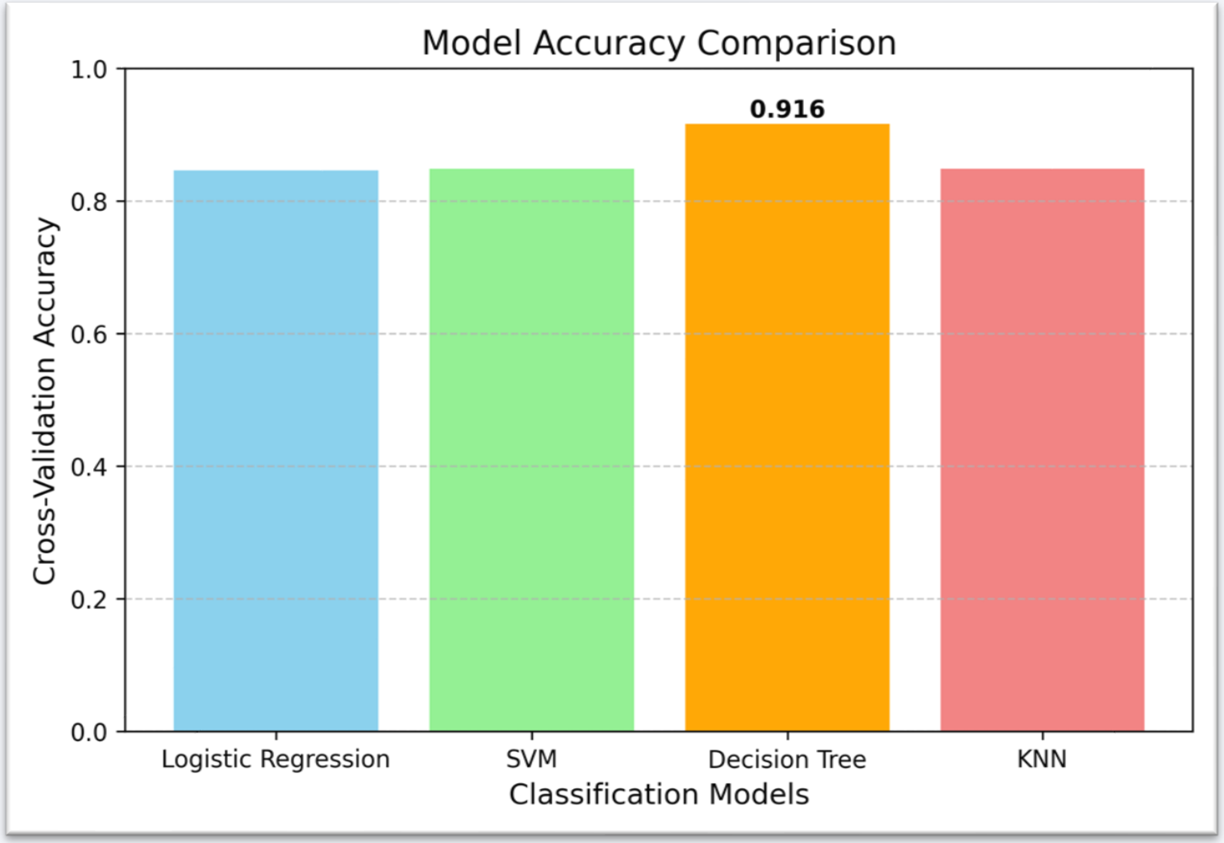
Project By: Shehraz Sarwar

Classification Accuracy

This bar chart compares the **cross-validation accuracies** of four classification models **Logistic Regression, SVM, Decision Tree,** and **KNN**.

Each bar represents how well a model performed during validation, with accuracy shown on the Y-axis.

Among all models, the **Decision Tree Classifier** achieved the **highest accuracy of 91.6%** , making it the **best-performing model** for predicting SpaceX launch success.



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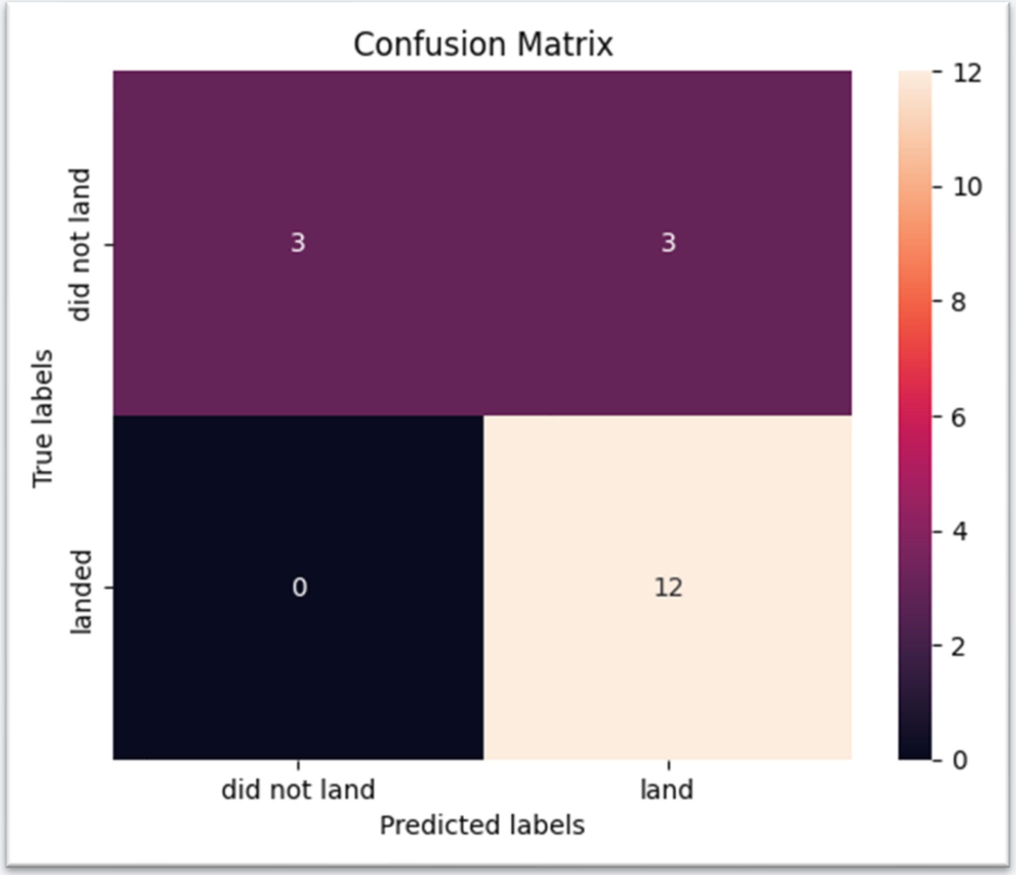
Confusion Matrix

The confusion matrix illustrates the performance of the Decision Tree model in classifying rocket landing outcomes.

- The model **correctly predicted 3 cases** where rockets did not land and **12 cases** where rockets landed successfully.
- It made **3 false positives** (predicted a landing when it did not happen) and **0 false negatives** (missed actual landings).

Overall, the results indicate **high accuracy**, as most predictions fall along the diagonal, demonstrating the model's strong ability to correctly identify successful landings.



Conclusions

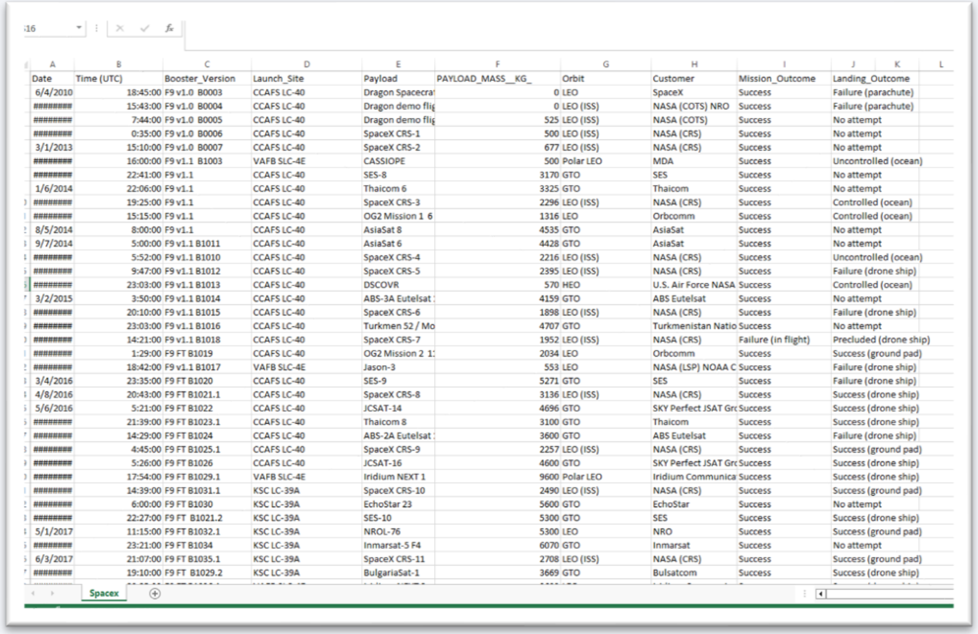
- The project successfully analyzed **SpaceX launch data** to predict rocket landing outcomes.
- Covered the **complete data science workflow** from **data gathering** to **EDA and visualization (SQL & Python)**, followed by an **interactive dashboard (Dash)** and **machine learning modeling**.
- Multiple classification models were trained and compared using **cross-validation accuracy**.
- Among all models, the **Decision Tree Classifier** achieved the **highest performance (91.6%)**.
- This demonstrates the model's strong ability to capture complex launch patterns and predict successful landings accurately.
- Overall, this project highlights the **integration of analytics, visualization, and AI** to support **data-driven decisions in space exploration**.

Project By: Shehraz Sarwar

Appendix

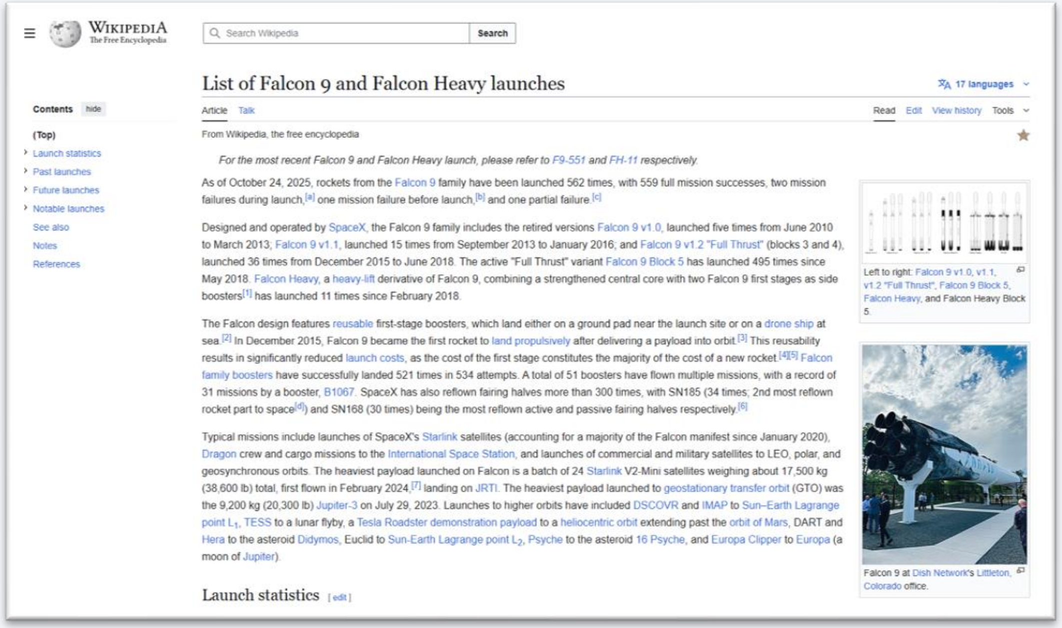
A. Data Sources

- SpaceX Launch Data (IBM Data Science Capstone)
- Wikipedia – Falcon 9 & Falcon Heavy launch details
- SQL database used for structured queries



A	B	C	D	E	F	G	H	I	J	K	L
Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome		
6/4/2010	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecra	0	LEO	SpaceX	Success	Failure (parachute)		
########	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flig	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)		
########	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flig	525	LEO (ISS)	NASA (COTS)	Success	No attempt		
########	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt		
3/1/2013	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt		
########	16:00:00	F9 v1.1 B1003	VAFB SLC-4E	CASSIOPE	500	Polar LEO	MDA	Success	Uncontrolled (ocean)		
########	22:41:00	F9 v1.1	CCAFS LC-40	SES-8	3170	GTO	SES	Success	No attempt		
1/6/2014	22:06:00	F9 v1.1	CCAFS LC-40	Thaicom 6	3325	GTO	Thaicom	Success	No attempt		
########	19:25:00	F9 v1.1	CCAFS LC-40	SpaceX CRS-3	2296	LEO (ISS)	NASA (CRS)	Success	Controlled (ocean)		
########	15:15:00	F9 v1.1	CCAFS LC-40	OG2 Mission 1 6	1316	LEO	Orbcomm	Success	Controlled (ocean)		
8/5/2014	8:00:00	F9 v1.1	CCAFS LC-40	AsiaSat 8	4535	GTO	AsiaSat	Success	No attempt		
9/7/2014	5:00:00	F9 v1.1 B1011	CCAFS LC-40	AsiaSat 6	4428	GTO	AsiaSat	Success	No attempt		
########	5:52:00	F9 v1.1 B1010	CCAFS LC-40	SpaceX CRS-4	2216	LEO (ISS)	NASA (CRS)	Success	Uncontrolled (ocean)		
########	9:47:00	F9 v1.1 B1012	CCAFS LC-40	SpaceX CRS-5	2395	LEO (ISS)	NASA (CRS)	Success	Failure (drone ship)		
########	23:03:00	F9 v1.1 B1013	CCAFS LC-40	DSCOVR	570	HEO	U.S. Air Force NASA	Success	Controlled (ocean)		
3/2/2015	3:50:00	F9 v1.1 B1014	CCAFS LC-40	ABS-3A Eutelsat	4159	GTO	ABS Eutelsat	Success	No attempt		
########	20:10:00	F9 v1.1 B1015	CCAFS LC-40	SpaceX CRS-6	1898	LEO (ISS)	NASA (CRS)	Success	Failure (drone ship)		
########	23:03:00	F9 v1.1 B1016	CCAFS LC-40	Turkmen 52 / Mo	4707	GTO	Turkmenistan Natio	Success	No attempt		
########	14:21:00	F9 v1.1 B1018	CCAFS LC-40	SpaceX CRS-7	1952	LEO (ISS)	NASA (CRS)	Failure (in flight)	Precluded (drone ship)		
########	1:29:00	F9 FT B1019	CCAFS LC-40	OG2 Mission 2 1	2034	LEO	Orbcomm	Success	Success (ground pad)		
########	18:42:00	F9 v1.1 B1017	VAFB SLC-4E	Jason-3	553	LEO	NASA (LSP) NOAA C	Success	Failure (drone ship)		
3/4/2016	23:35:00	F9 FT B1020	CCAFS LC-40	SES-9	5271	GTO	SES	Success	Failure (drone ship)		
4/8/2016	20:43:00	F9 FT B1021.1	CCAFS LC-40	SpaceX CRS-8	3136	LEO (ISS)	NASA (CRS)	Success	Success (drone ship)		
5/6/2016	5:21:00	F9 FT B1022	CCAFS LC-40	JCSAT-14	4696	GTO	SKY Perfect JSAT Gr	Success	Success (drone ship)		
########	21:39:00	F9 FT B1023.1	CCAFS LC-40	Thaicom 8	3100	GTO	Thaicom	Success	Success (drone ship)		
########	14:29:00	F9 FT B1024	CCAFS LC-40	ABS-2A Eutelsat	3600	GTO	ABS Eutelsat	Success	Failure (drone ship)		
########	4:45:00	F9 FT B1025.1	CCAFS LC-40	SpaceX CRS-9	2257	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)		
########	5:26:00	F9 FT B1026	CCAFS LC-40	JCSAT-16	4600	GTO	SKY Perfect JSAT Gr	Success	Success (drone ship)		
########	17:54:00	F9 FT B1029.1	VAFB SLC-4E	Iridium NEXT 1	9600	Polar LEO	Iridium Communica	Success	Success (drone ship)		
########	14:39:00	F9 FT B1031.1	KSC LC-39A	SpaceX CRS-10	2490	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)		
########	6:00:00	F9 FT B1030	KSC LC-39A	EchoStar 23	5600	GTO	EchoStar	Success	No attempt		
########	22:27:00	F9 FT B1021.2	KSC LC-39A	SES-10	5300	GTO	SES	Success	Success (drone ship)		
5/1/2017	11:15:00	F9 FT B1032.1	KSC LC-39A	NROL-76	5300	LEO	NRO	Success	Success (ground pad)		
########	23:21:00	F9 FT B1034	KSC LC-39A	Inmarsat-5 F4	6070	GTO	Inmarsat	Success	No attempt		
6/3/2017	21:07:00	F9 FT B1035.1	KSC LC-39A	SpaceX CRS-11	2708	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)		
########	19:10:00	F9 FT B1029.2	KSC LC-39A	BulgariaSat-1	3669	GTO	Bulsatcom	Success	Success (drone ship)		

Spacex



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List of Falcon 9 and Falcon Heavy launches

17 languages

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From Wikipedia, the free encyclopedia

For the most recent Falcon 9 and Falcon Heavy launch, please refer to F9-551 and FH-11 respectively.

As of October 24, 2025, rockets from the Falcon 9 family have been launched 562 times, with 559 full mission successes, two mission failures during launch,[a] one mission failure before launch,[b] and one partial failure.[c]

Designed and operated by SpaceX, the Falcon 9 family includes the retired versions Falcon 9 v1.0, launched five times from June 2010 to March 2013; Falcon 9 v1.1, launched 15 times from September 2013 to January 2016; and Falcon 9 v1.2 "Full Thrust" (blocks 3 and 4), launched 36 times from December 2015 to June 2018. The active "Full Thrust" variant Falcon 9 Block 5 has launched 495 times since May 2018. Falcon Heavy, a heavy-lift derivative of Falcon 9, combining a strengthened central core with two Falcon 9 first stages as side boosters[1] has launched 11 times since February 2018.

Left to right: Falcon 9 v1.0, v1.1, v1.2 "Full Thrust", Falcon 9 Block 5, Falcon Heavy, and Falcon Heavy Block 5.

The Falcon design features reusable first-stage boosters, which land either on a ground pad near the launch site or on a drone ship at sea.[2] In December 2015, Falcon 9 became the first rocket to land propulsively after delivering a payload into orbit.[3] This reusability results in significantly reduced launch costs, as the cost of the first stage constitutes the majority of the cost of a new rocket.[4][5] Falcon family boosters have successfully landed 521 times in 534 attempts. A total of 51 boosters have flown multiple missions, with a record of 31 missions by a booster, B1067. SpaceX has also reflown fairing halves more than 300 times, with SN185 (34 times; 2nd most reflown rocket part to space[d]) and SN168 (30 times) being the most reflown active and passive fairing halves respectively.[6]

Typical missions include launches of SpaceX's Starlink satellites (accounting for a majority of the Falcon manifest since January 2020), Dragon crew and cargo missions to the International Space Station, and launches of commercial and military satellites to LEO, polar, and geosynchronous orbits. The heaviest payload launched on Falcon is a batch of 24 Starlink V2-Mini satellites weighing about 17,500 kg (38,600 lb) total, first flown in February 2024,[7] landing on JRTI. The heaviest payload launched to geostationary transfer orbit (GTO) was the 9,200 kg (20,300 lb) Jupiter-3 on July 29, 2023. Launches to higher orbits have included DSCOVR and IMAP to Sun–Earth Lagrange point L_1, TESS to a lunar flyby, a Tesla Roadster demonstration payload to a heliocentric orbit extending past the orbit of Mars, DART and Hera to the asteroid Didymos, Euclid to Sun-Earth Lagrange point L_2, Psyche to the asteroid 16 Psyche, and Europa Clipper to Europa (a moon of Jupiter).

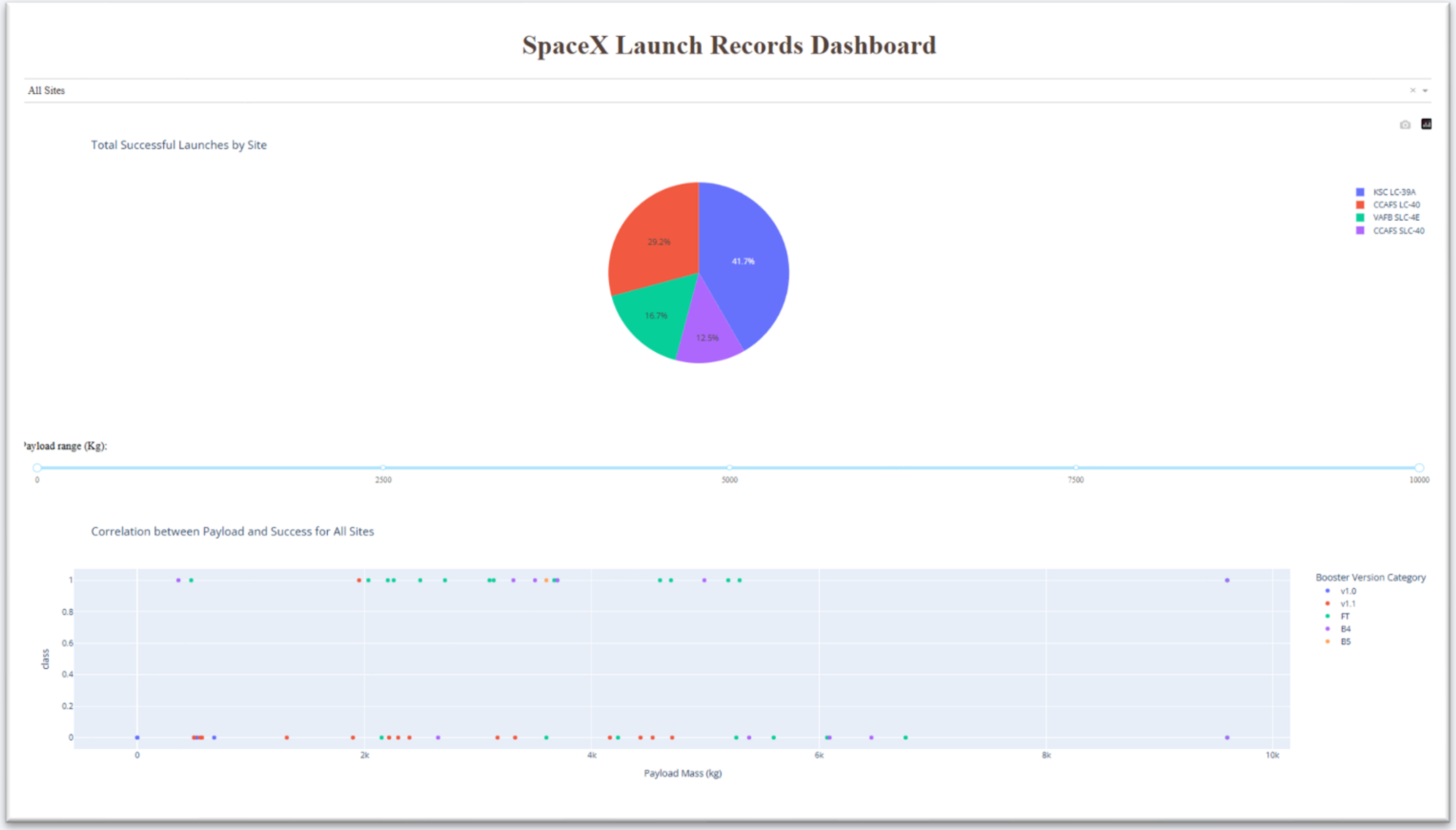
Falcon 9 at Dish Network's Littleton, Colorado office.

Launch statistics [edit]

Appendix

B. Dash App Snapshot

- Built with **Plotly Dash**
- Pie chart, Dropdown, Scatter & Trend chart



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Appendix

C. References

- IBM Applied Data Science Capstone Project:
  https://www.coursera.org/learn/applied-data-science-capstone
- Scikit-learn Docs: https://scikit-learn.org/stable/
- Plotly Dash Docs: https://dash.plotly.com/
- Wikipedia – *Falcon 9 & Falcon Heavy launches:*
  https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches

Thank you!

<u>My LinkedIn</u>: https://www.linkedin.com/in/shehraz-sarwar-ghouri-321394247/

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